Combining Machine-Learning and Dynamic Network Models for Sepsis Prediction

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Sepsis and the Dynamic Network Model

Sepsis and the Dynamic Network Model

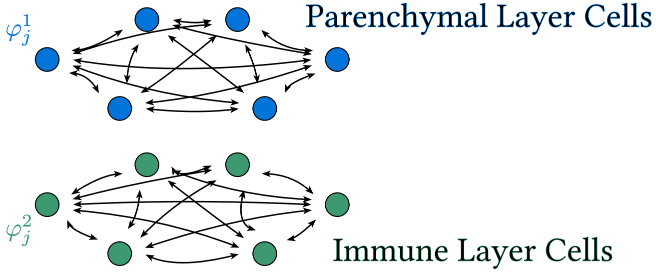
Functional modeling (adaptive cell communication)

- Interaction of parenchymal (organ) and immune cells via cytokines

Two-layer dynamic network model

- Adaptive phase oscillators

Dynamic Network Model (DNM)



$$\dot{\varphi}_i^1 = -\frac{1}{N}\sum_{j=1}^{N}\left\{\left(1+\kappa_{ij}^1\right)\sin\left(\varphi_i^1 - \varphi_j^1 - 0.28\pi\right)\right\} - \sigma\sin\left(\varphi_i^1 - \varphi_i^2\right)$$

$$\dot{\kappa}_{ij}^1 = -0.03\left(\kappa_{ij}^1 + \sin\left(\varphi_i^1 - \varphi_j^1 - \beta\right)\right)$$

$$\dot{\varphi}_i^2 = -\frac{1}{N}\sum_{j=1}^{N}\left\{\kappa_{ij}^2\sin\left(\varphi_i^2 - \varphi_j^2 - 0.28\pi\right)\right\} - \sigma\sin\left(\varphi_i^2 - \varphi_i^1\right)$$

$$\dot{\kappa}_{ij}^2 = -0.3\left(\kappa_{ij}^2 + \sin\left(\varphi_i^2 - \varphi_j^2 - \beta\right)\right)$$

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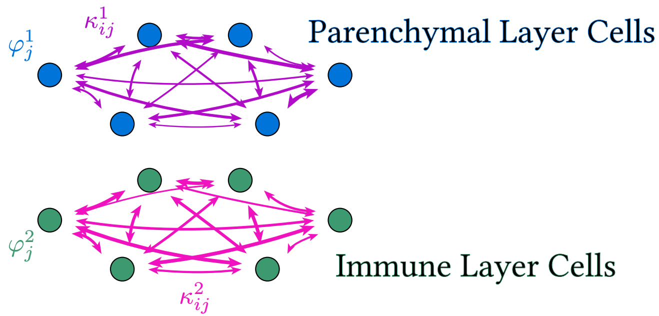

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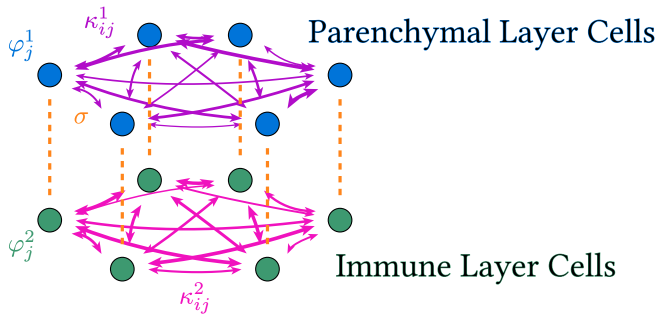

$$\dot{\varphi}^1_i = -\frac{1}{N}\sum_{j=1}^{N}\left\{\left(1+\kappa^1_{ij}\right)\sin\left(\varphi^1_i - \varphi^1_j - 0.28\pi\right)\right\} - \sigma\sin\left(\varphi^1_i - \varphi^2_i\right)$$

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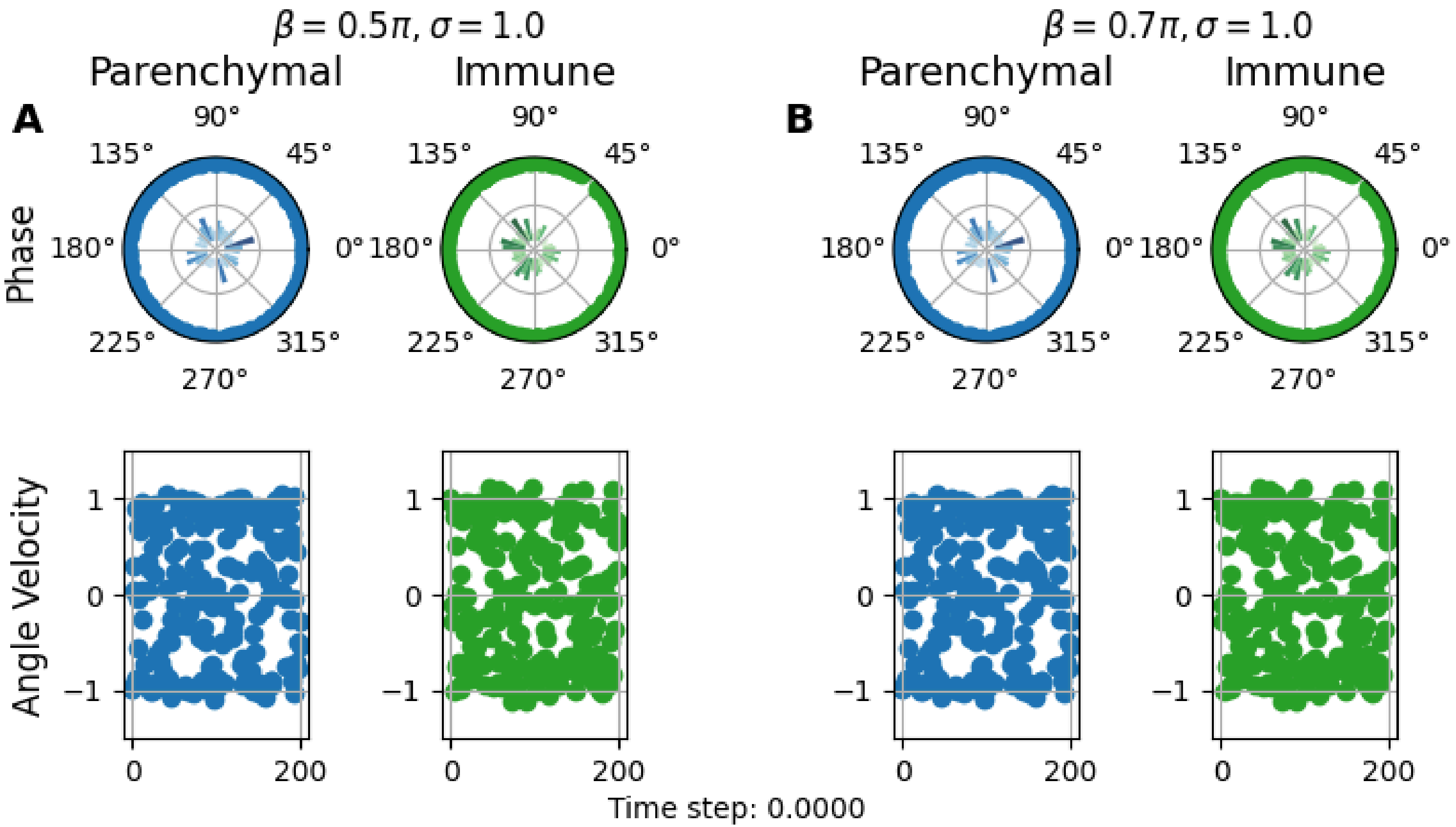
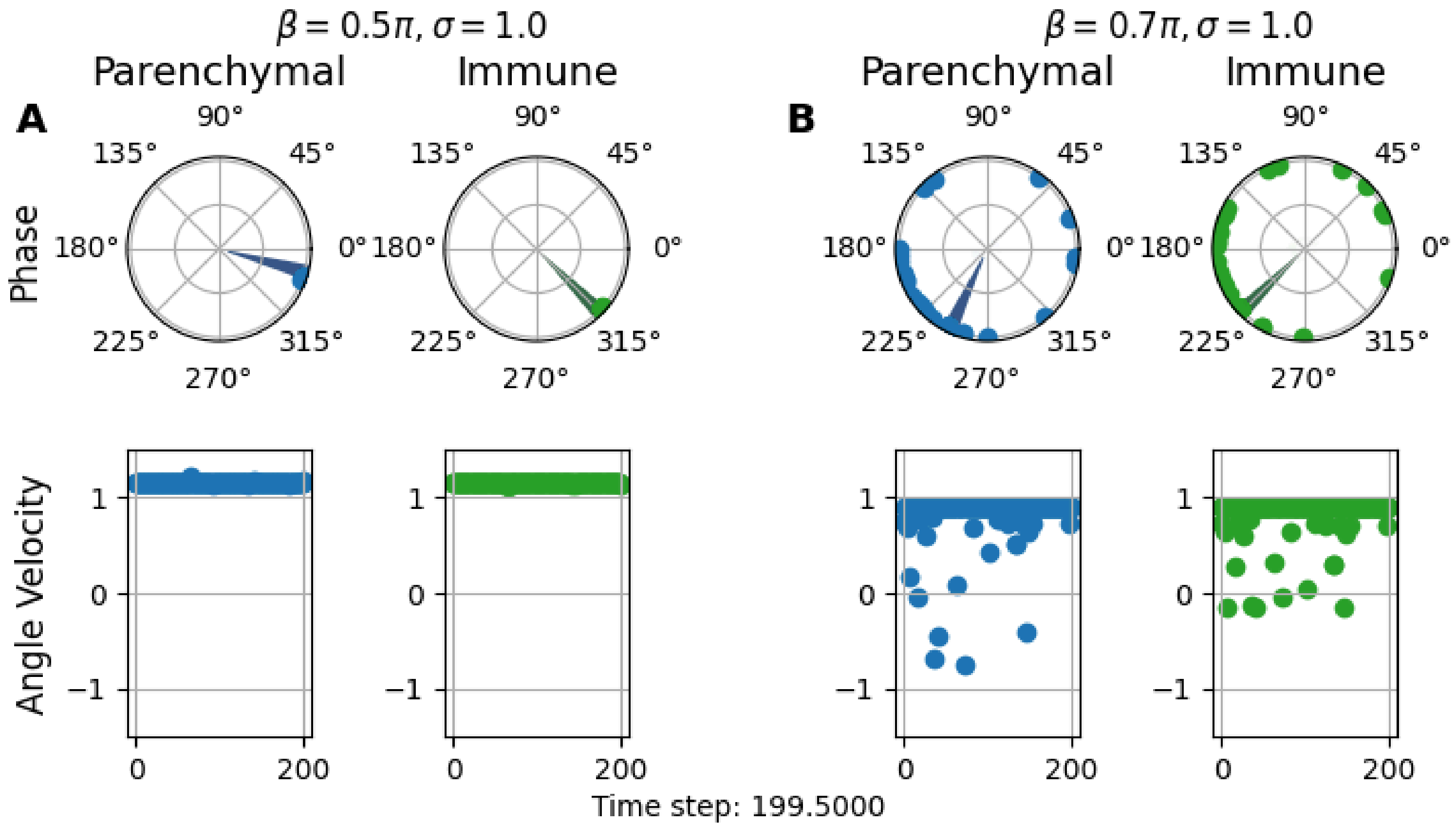


Sepsis and the Dynamic Network Model

Simulation Video

Sepsis and the Dynamic Network Model



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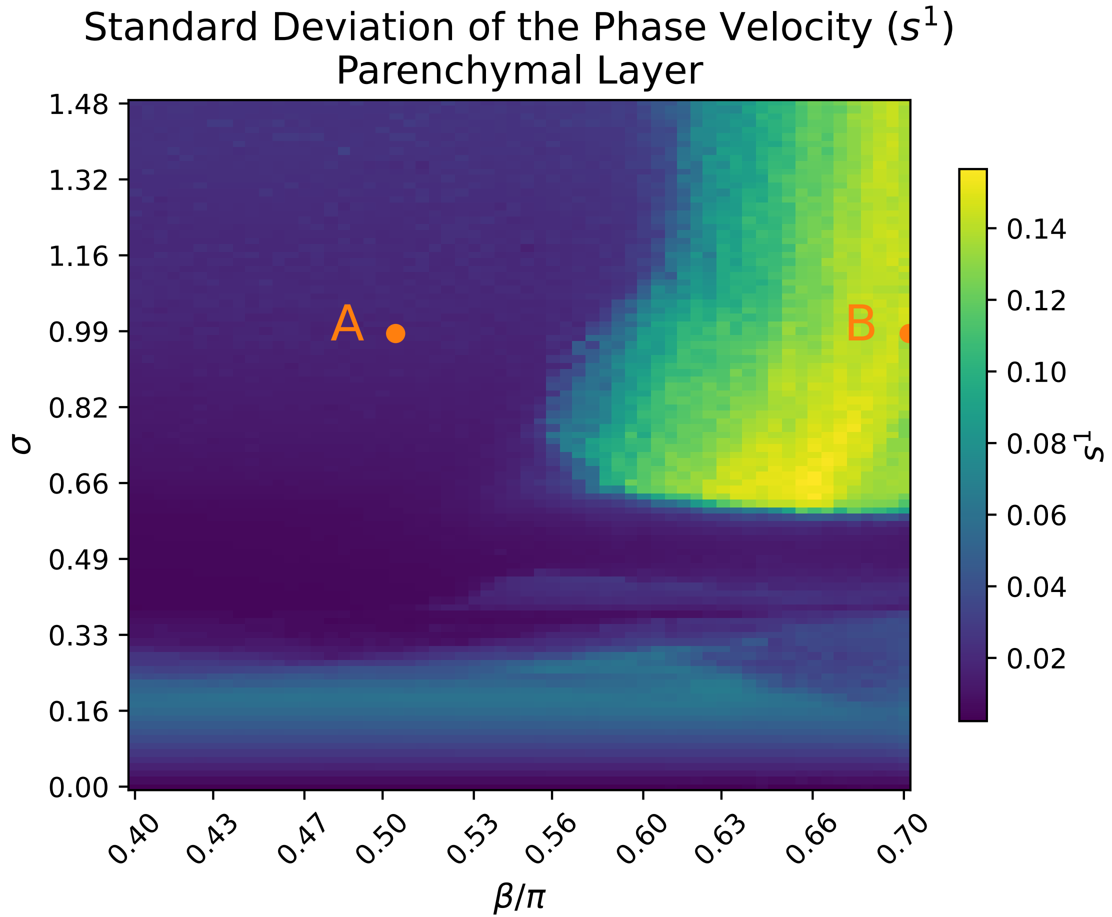
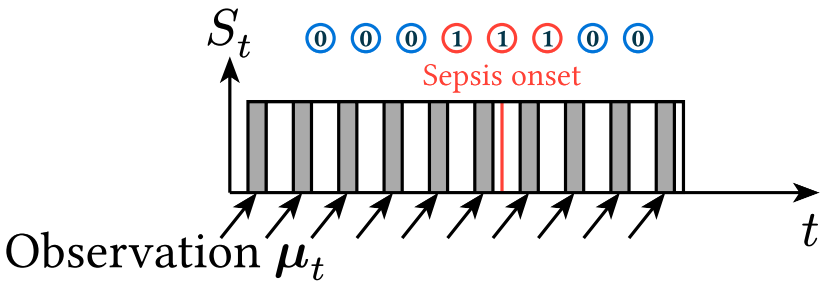


Latent Dynamics Model

Latent Dynamics Model

Online Prediction

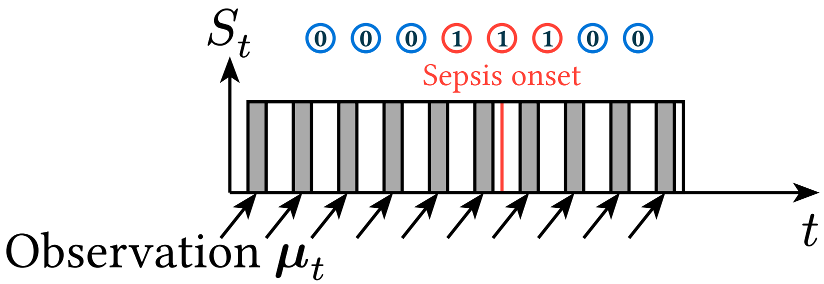


Sepsis Definition (simplified) $\rightarrow S_t$

1. Infection $\rightarrow I_t$
2. Rapid Organ Failure $\rightarrow A_t$
    - SOFA-score increase ≥ 2

Latent Dynamics Model

Online Prediction



$\boldsymbol{\mu}_t$

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Sepsis Risk
$\tilde{S}_t = \tilde{A}_t \tilde{I}_t$

Latent Dynamics Model

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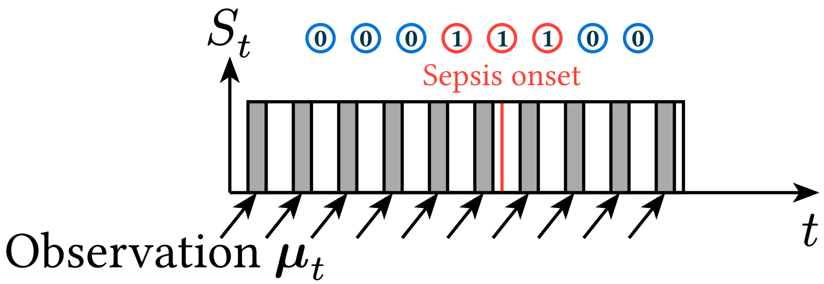

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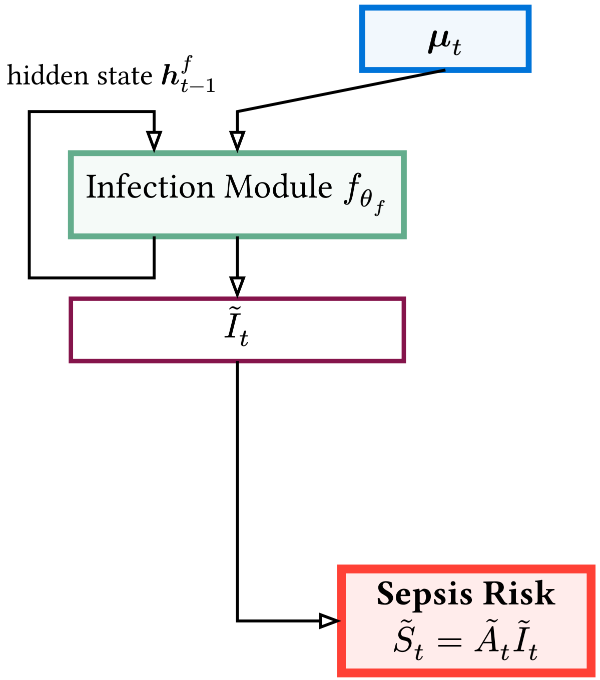

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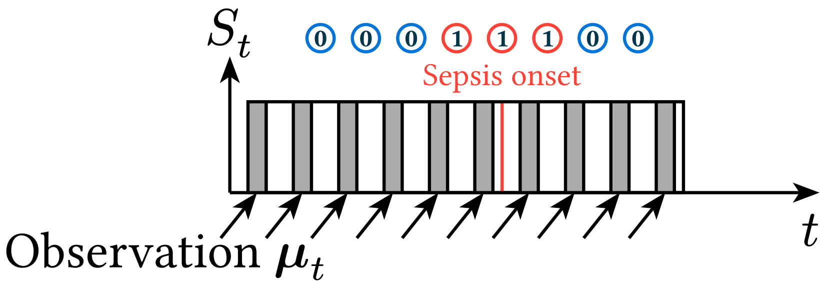

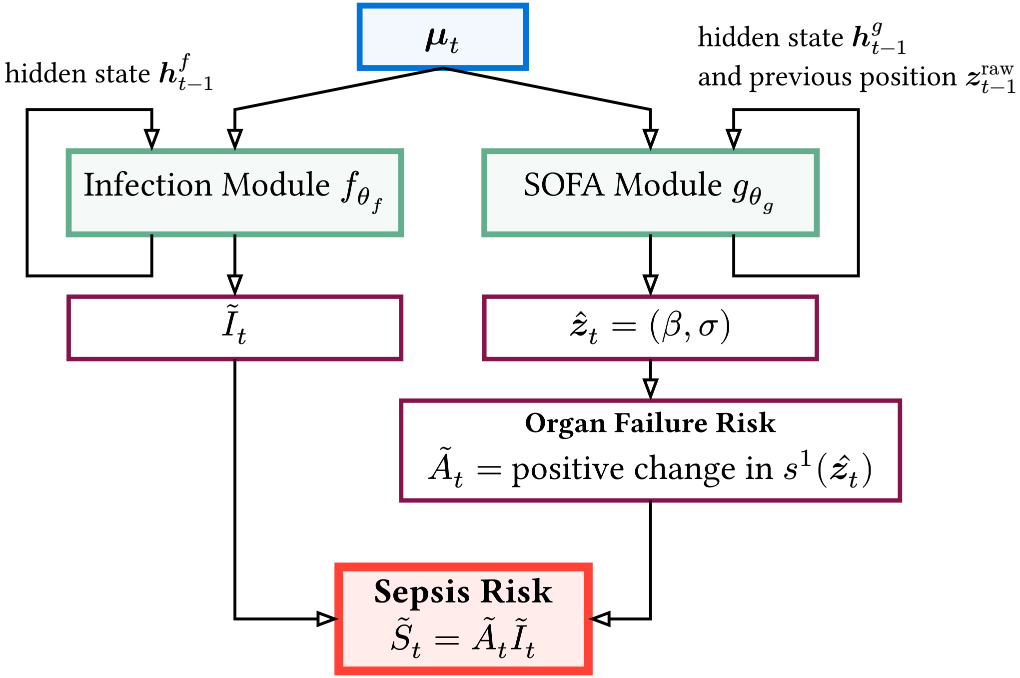

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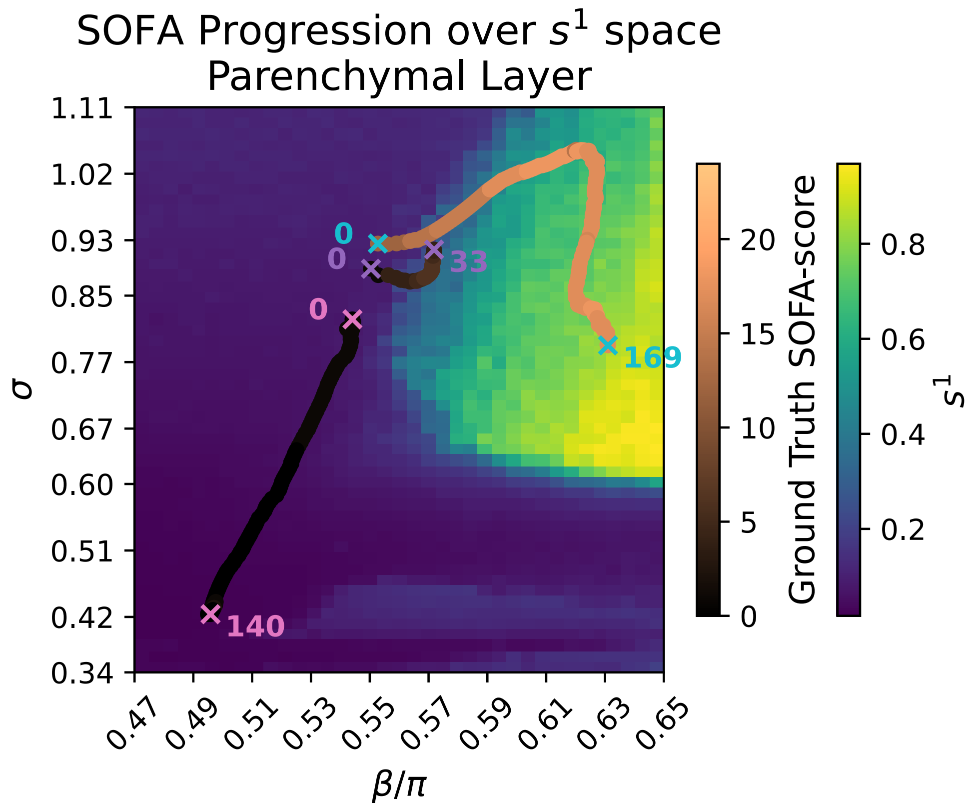
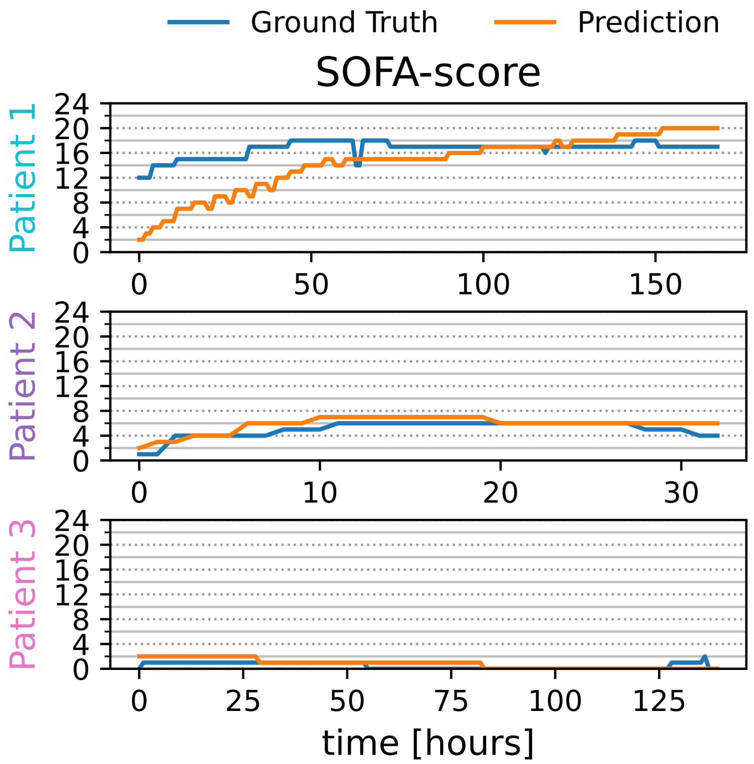
Experimental Results

Experimental Results





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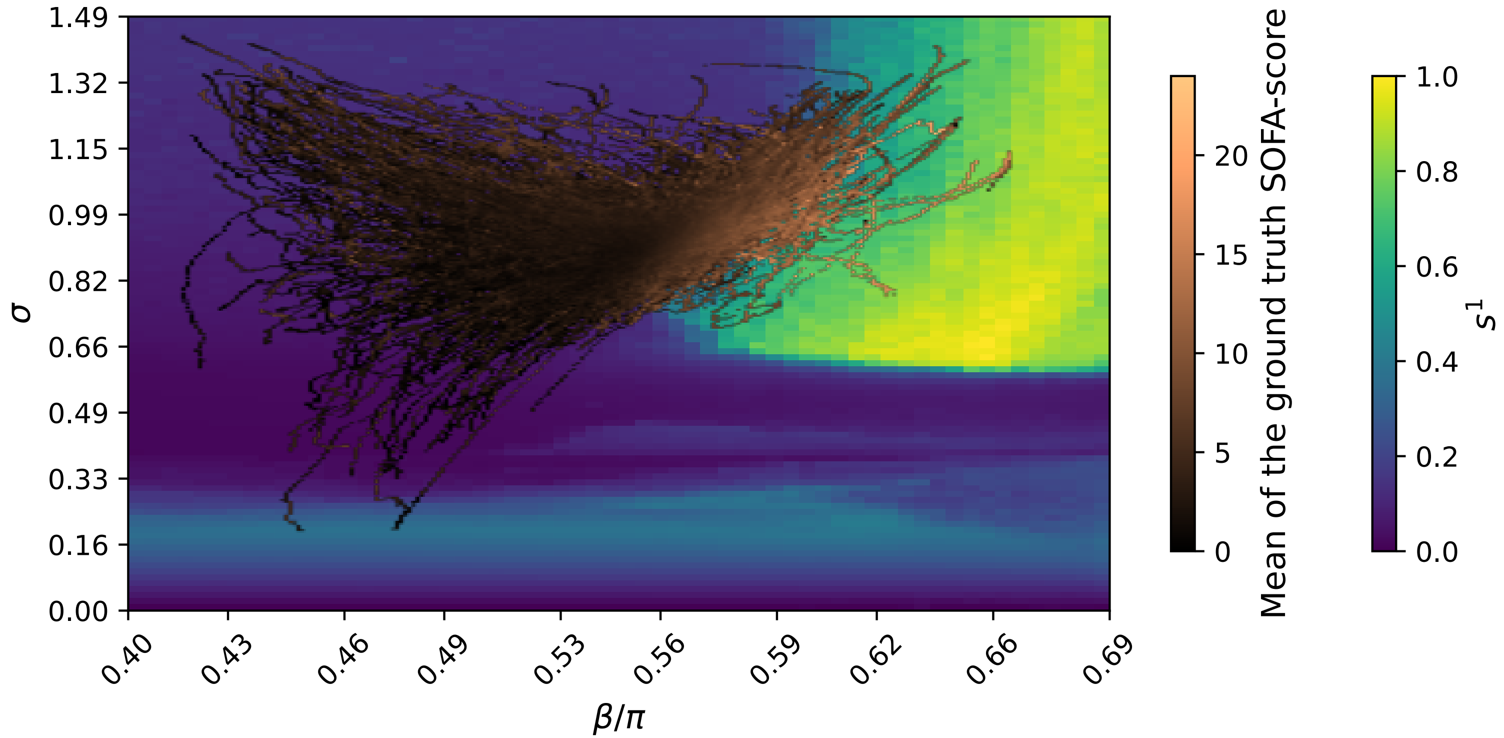


Experimental Results

Comparison against mean performance of YAIB [1],
AUROC $\times$ 100 ($\uparrow$, higher is better) and AUPRC $\times$ 100 ($\uparrow$).

Model	**AUROC**	**AUPRC**
YAIB		
Regularized Logistic Regression	77.1 $\pm$(0.4)	4.6 $\pm$(0.1)
Light Gradient Boosting Machine	77.5 $\pm$(0.3)	5.9 $\pm$(0.2)
Transformer	80.0 $\pm$(0.8)	6.6 $\pm$(0.2)
LSTM	82.0 $\pm$(0.3)	8.0 $\pm$(0.2)
Temporal Convolutional Network	82.7 $\pm$(0.3)	8.8 $\pm$(0.2)
GRU	**83.6** $\pm$(0.3)	**9.1** $\pm$(0.3)
This work		
Latent Dynamics Model	**84.01** $\pm$(0.8)	**9.97** $\pm$(0.8)

Conclusion

Conclusion

- Interpretable
- Decomposition Strategy
- Competitive Performance

- Do other sepsis models exist, that work even better?

https://github.com/unartig/sepsis_osc

Bibliography

Bibliography

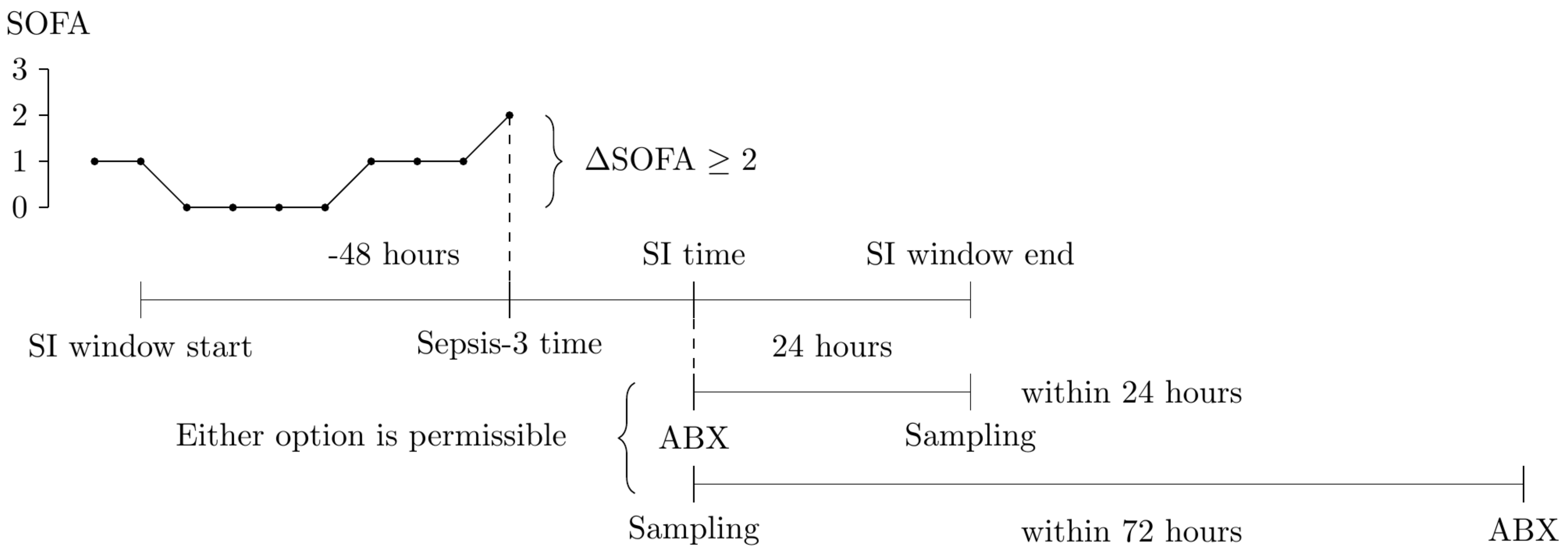
[1] R. van de Water, H. N. A. Schmidt, P. Elbers, P. Thoral, B. Arnrich, and P. Rockenschaub, "Yet Another ICU Benchmark: A Flexible Multi-Center Framework for Clinical ML," in *The Twelfth International Conference on Learning Representations*, Oct. 2024.

[2] N. Bennett, D. Plečko, and I.-F. Ukor, "Sepsis 3 label — sep3." [Online]. Available: https://eth-mds.github.io/ricu/reference/label_sep3.html

Supplemental Material

Supplemental Material

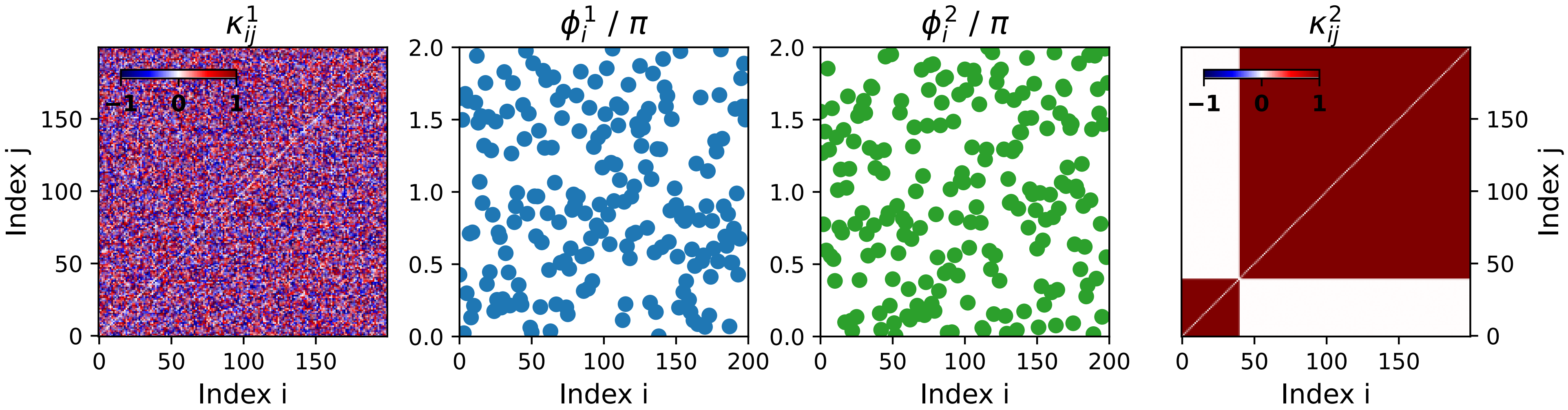
Graphical representation of the timings in the Sepsis-3 definition, taken from [2].



Supplemental Material

Initialization

- Uniform cytokine activity in the organ layer
- Uniform initial phases
- Cytokine activation clusters in the immune layer



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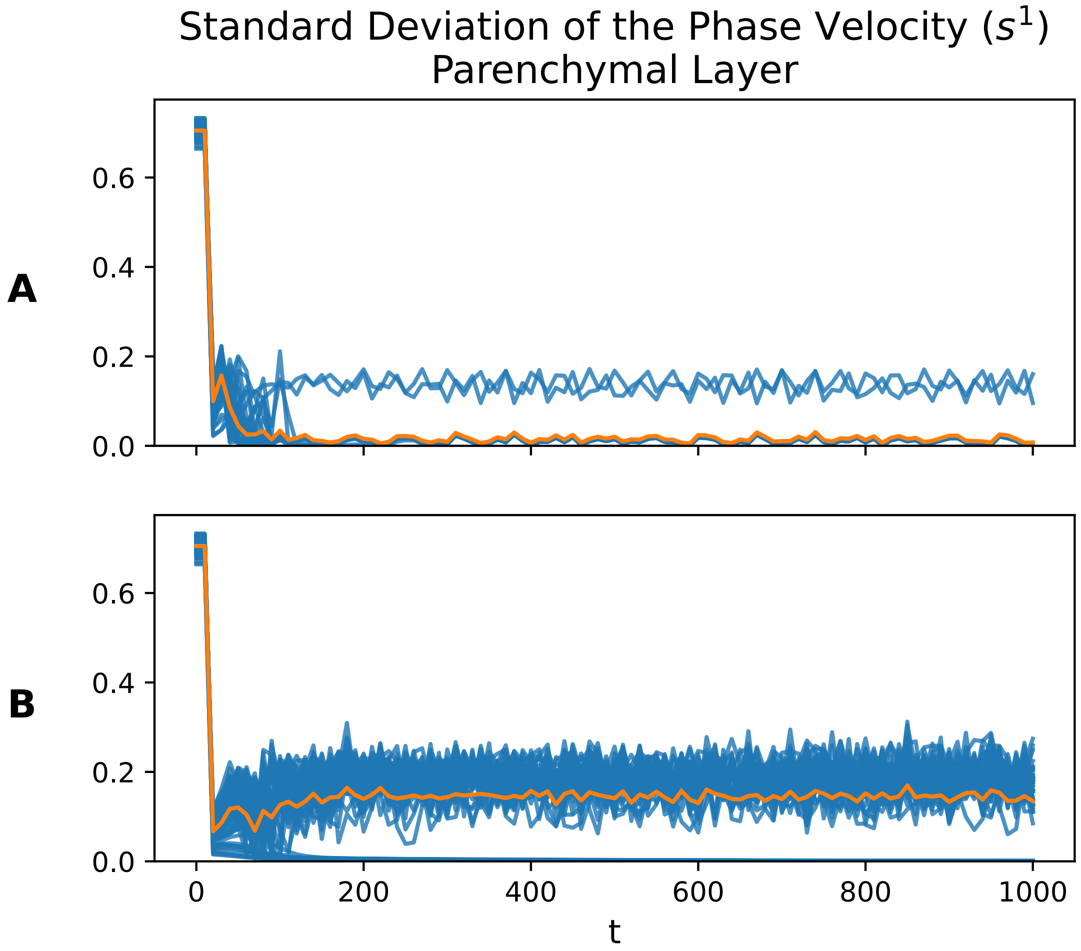

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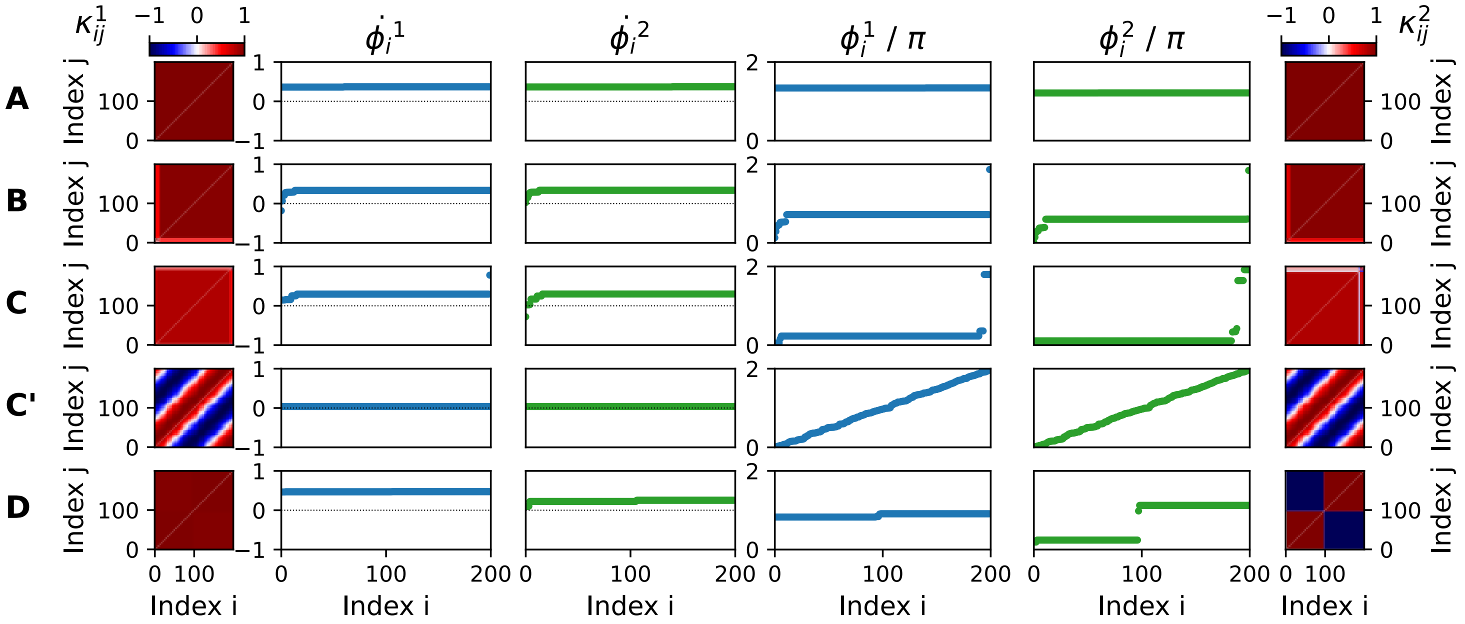

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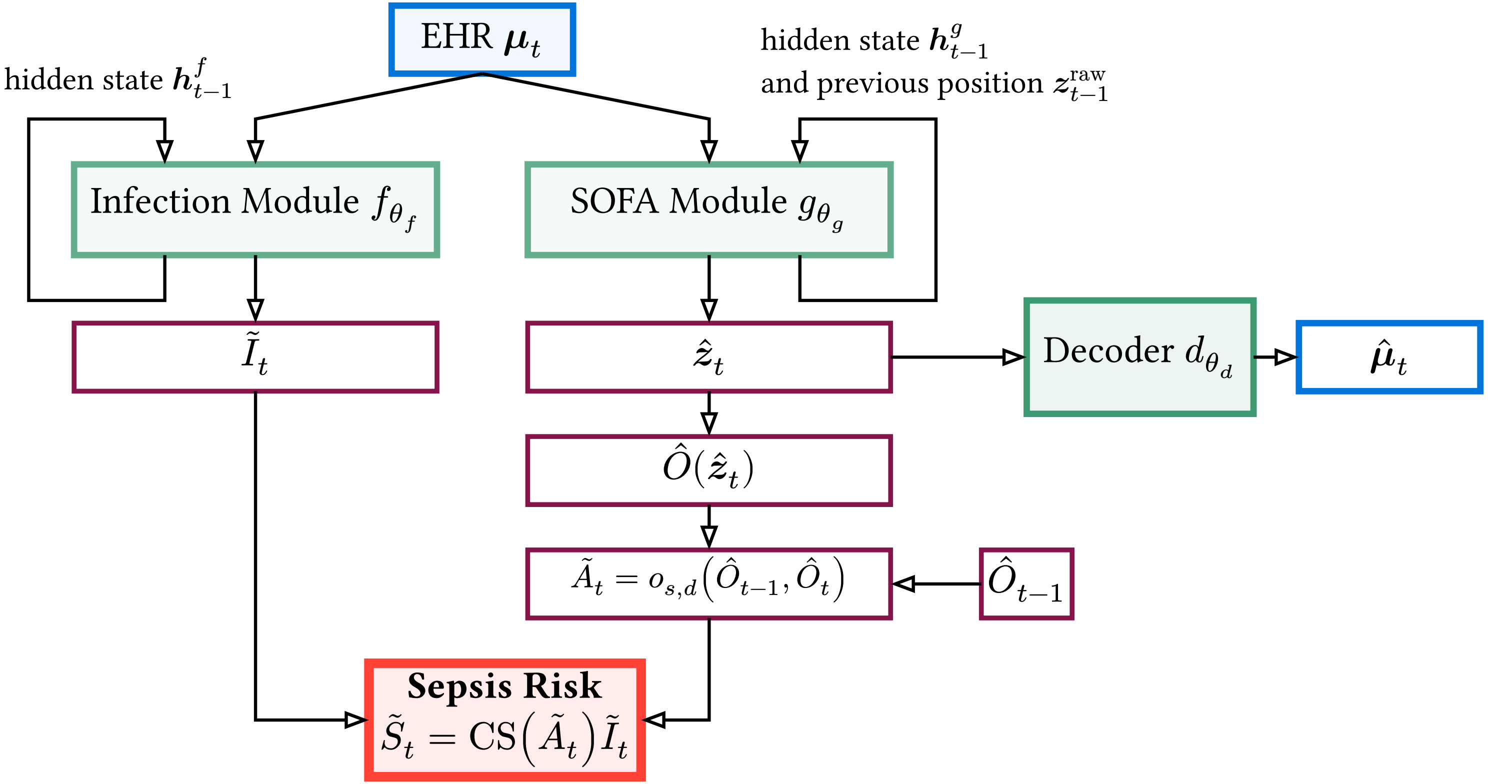

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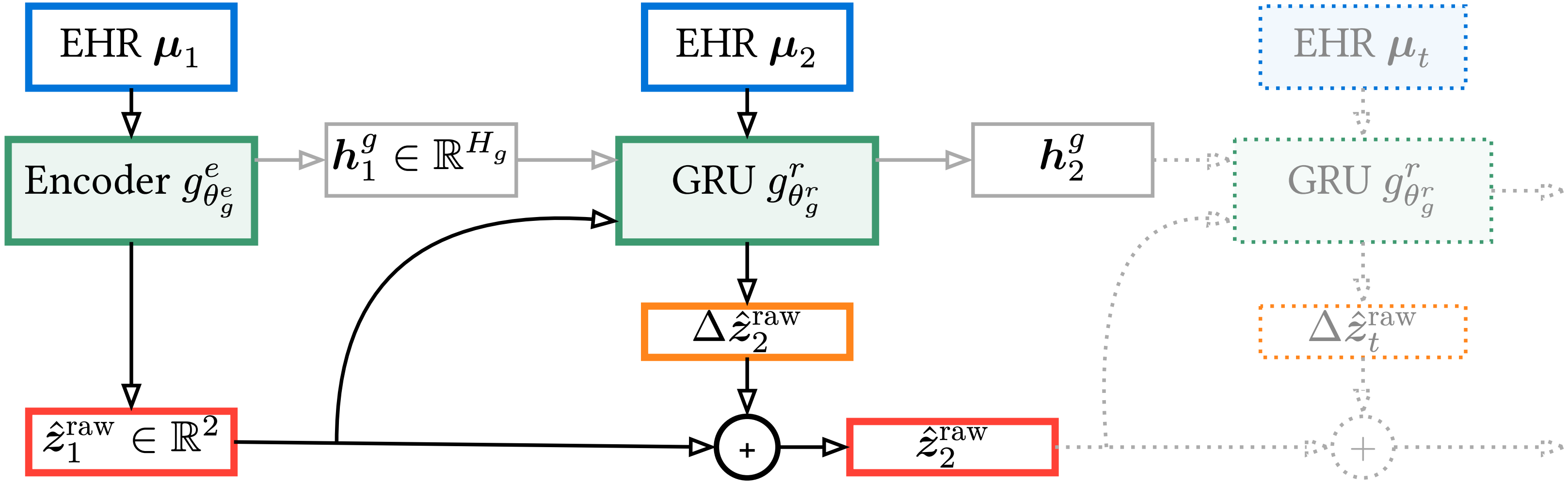

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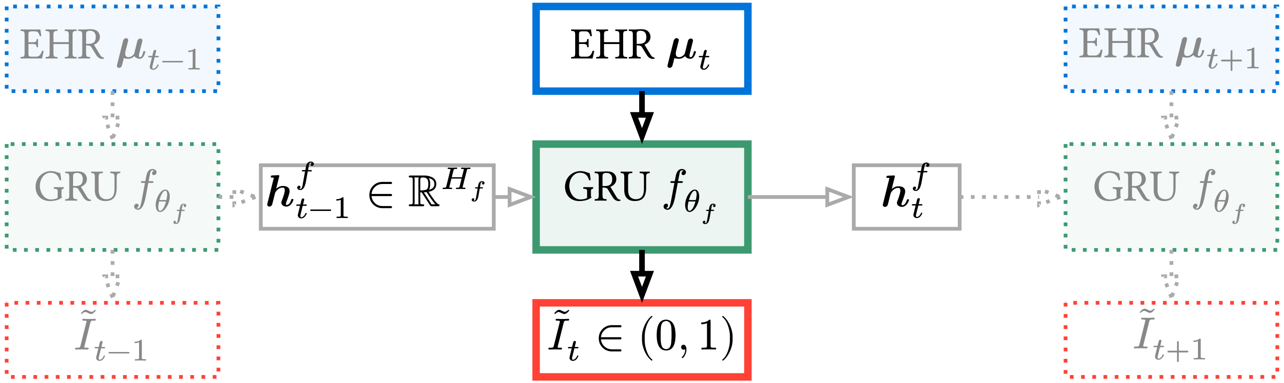

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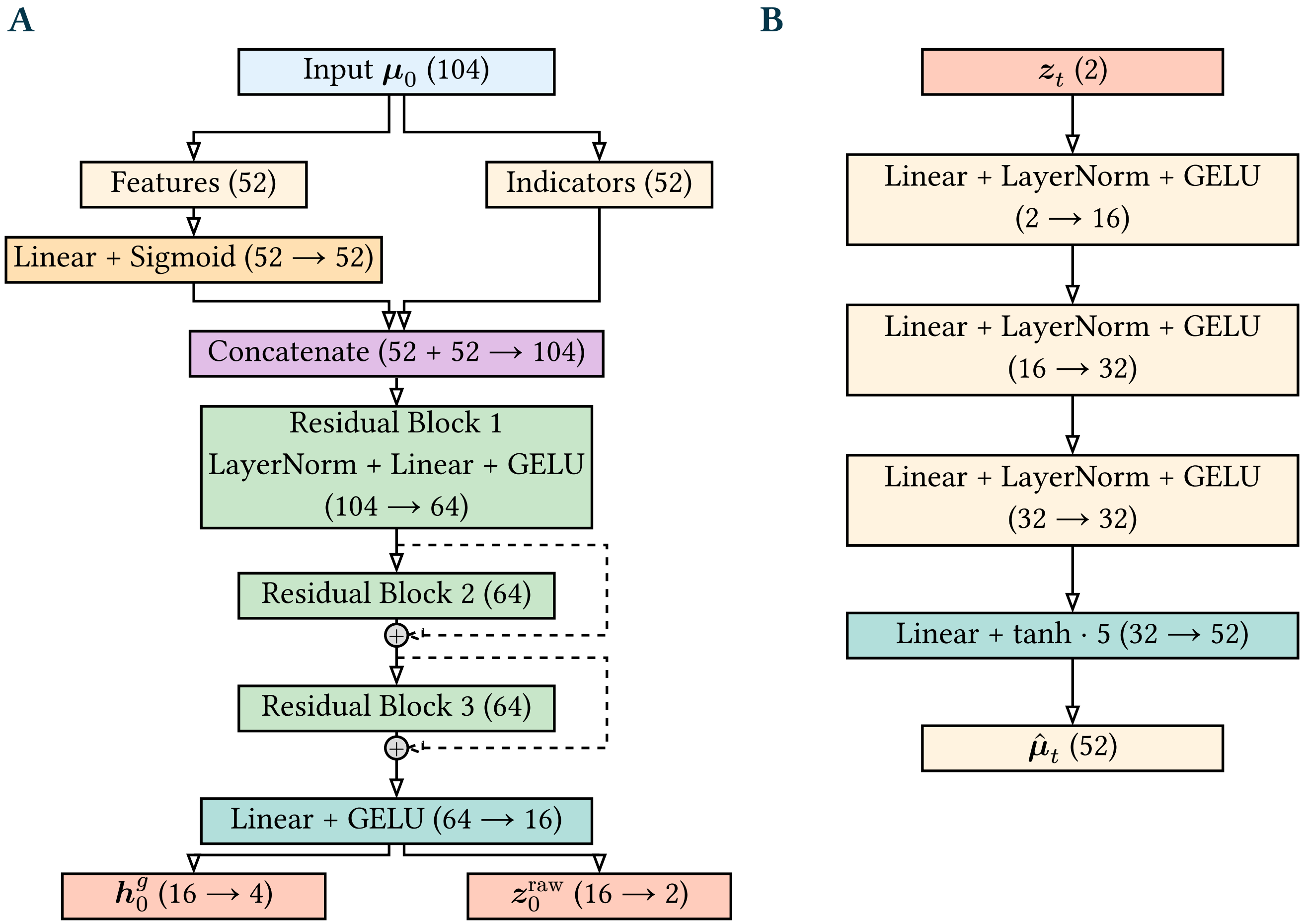

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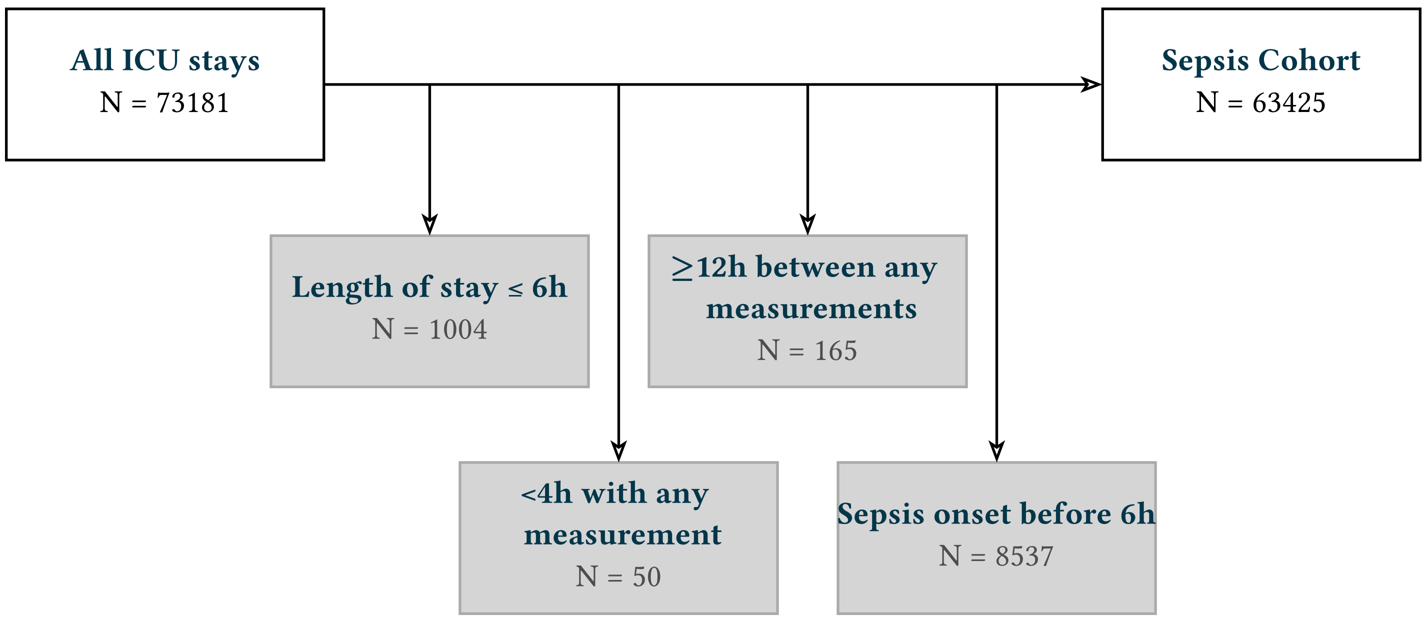


Supplemental Material

Characteristic	All patients	SEP-3 positive	SEP-3 negative
Demographics			
N	63425 (100.0)	3320 (5.2)	60105 (94.8)
Male	35170 (55.5)	1881 (56.7)	33289 (55.4)
Age at admission	65.0 (53.0–76.0)	65.0 (54.0–76.0)	65.0 (53.0–76.0)
Weight at admission	77.6 (65.1–92.3)	77.6 (65.6–94.0)	77.6 (65.0–92.2)

Supplemental Material

Characteristic	All patients	SEP-3 positive	SEP-3 negative
Clinical Outcomes			
SOFA median	3.0 (1.0–5.0)	3.0 (1.0–5.0)	3.0 (1.0–5.0)
SOFA max	4.0 (2.0–6.0)	5.0 (4.0–8.0)	4.0 (2.0–6.0)
Hospital LOS hours	157.7 (92.8–268.9)	335.1 (194.2–548.6)	150.3 (90.9–256.0)
Hospital Mortality	4828 (7.6)	879 (26.5)	3949 (6.6)
Sepsis-3 onset time	-	13.0 (8.0–34.0)	-

Supplemental Material

Characteristic	All patients	SEP-3 positive	SEP-3 negative
Ethnicity			
White	40364 (63.6)	2087 (62.9)	38277 (63.7)
Black	5809 (9.2)	262 (7.9)	5547 (9.2)
Asian	721 (1.1)	42 (1.3)	679 (1.1)
Hispanic	630 (1.0)	32 (1.0)	598 (1.0)
Other/Unknown	14924 (23.5)	897 (27.0)	14027 (23.3)

Supplemental Material

Characteristic	All patients	SEP-3 positive	SEP-3 negative
Admission Type			
Medical	45009 (71.0)	2817 (84.8)	42192 (70.2)
Surgical	2239 (3.5)	45 (1.4)	2194 (3.7)
Other/Unknown	15200 (24.0)	458 (13.8)	14742 (24.5)

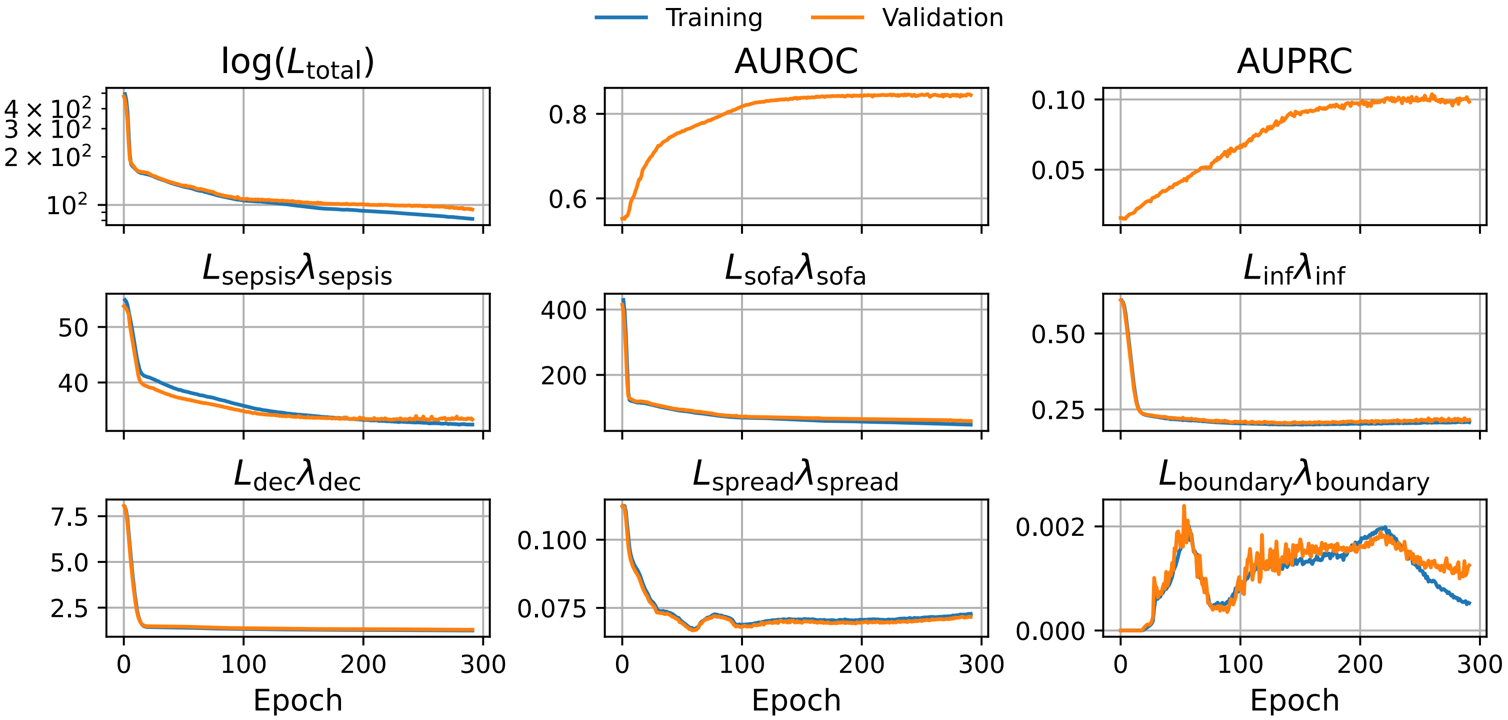
Supplemental Material

Overview of loss components in the LDM training objective.

Loss	**Type**	**Purpose**	**Supervises**
$\mathcal{L}_{\text{sepsis}}$	BCE	Primary sepsis prediction	$f_{\theta_f}, g^e_{\theta^e_g}, g^r_{\theta^r_g}$
$\mathcal{L}_{\text{sofa}}$	Weighted MSE	SOFA estimation	$g^e_{\theta^e_g}, g^r_{\theta^r_g}$
$\mathcal{L}_{\text{inf}}$	BCE	Infection indicator	f_{θ_f}
$\mathcal{L}_{\text{spread}}$	Covariance	Latent diversity	$g^e_{\theta^e_g}, g^r_{\theta^r_g}$
$\mathcal{L}_{\text{boundary}}$	Positional	Latent Space	$g^e_{\theta^e_g}, g^r_{\theta^r_g}$
$\mathcal{L}_{\text{dec}}$	MSE	Latent semantics	$d_{\theta_d}, (g^e_{\theta^e_g}, g^r_{\theta^r_g})$

Supplemental Material

Progression of training and validation losses



Supplemental Material

$$\delta\left(\tilde{S}_t\right) = \mathbb{I}\left(\tilde{S}_t > \tau\right)$$

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{TP} + \mathrm{FN}}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

Receiver Operating Characteristic and Precision-Recall Curve.